

Seattle Campus Secondhand Marketplace

Milestone: Project Proposal

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Problem Statement

Given the limited budget of university students, they buy and sell books, furniture, equipment, school supplies, and electronics among others. Physical marketplaces are geographically dispersed and at most times, incomplete. Virtual marketplaces also exist but they are general purpose and since they are mostly not affiliated with the university, safety and convenient exchanges may be challenging.

We are proposing a Seattle Campus Secondhand Marketplace. This is a database-driven online marketplace for colleges and universities in the Seattle area, with a pilot in Northeastern University - Seattle Campus.

The system will allow users to list secondhand items, filter and search, add to favorites, post interest and purchase requests, arrange for pick-up, and post reviews.

Administrators can manage categories, review listings and user reviews, regulate pick-up points, and analyze system usage activities.

Problem Definition

The project will design and implement a database system that will support the major activities of the proposed second hand marketplace

It aims to address the following questions:

- 1) How can listings, categories, users, transactions, pick-up locations, reviews be efficiently organized in a database?
- 2) How can students search and filter using categories, prices, location, condition and other attributes?
- 3) How can the system track the transaction cycle of an item from listing up to pick-up?
- 4) Which categories and price ranges have the highest activity, both in listing and purchases?
- 5) How can buyer, seller activity and their reviews be recorded?
- 6) What usage trends can help administrators better manage the system?

The resulting database will be used both for operational transactions as well as analytical activities to better manage the system.

Data Sources

A significant part of the data is envisioned to be created during actual operations, as users are added and their activities posted. However, for the purposes of this project, we will create them synthetically to test the database design and configurations that we will implement.

Sample databases that we will create will include students, items, categories, purchase requests, reviews, favorites and pickup locations

We will also use publicly available information, relating to locations, campuses and other potential public pick-up locations. We will properly cite this resources, as appropriate.

Data Description

The database contains around 10 major entity types.

Entity	Description	Example
User	A student buyer, seller, or platform administrator at a Seattle university.	UW computer science student Leo Wang sells a used monitor.
Item Category	A classification dictionary for listing and filtering goods.	Electronics, textbooks, home goods, outdoor gear, accessories.
Secondhand Item	A user-posted item available for sale.	A like-new Dell 27-inch monitor for \$80.

Transaction Order	A record of a completed buyer-seller agreement and its status.	Leo sells a monitor to Lily; pickup is pending.
Purchase Request	A buyer's request for a specific item.	A UW CSE sophomore seeks textbooks under \$200.
Campus Pickup Point	A campus location for in-person exchanges.	UW Red Square or the SU Student Center.
Item Tag	An item attribute used for search and recommendations.	Like new, negotiable, UW pickup, accessories included.
Transaction Review	A post-transaction rating between buyer and seller.	A five-star review: "Condition matched the description."
Favorite	A record of an item saved by a user.	Lily saves a used mechanical keyboard.
Report	A report of an inappropriate item or transaction.	A counterfeit-brand report marked received.